

Comparative study of generalized-sampling-theorem-based digital super-resolution for 2D data

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The size of pixels in a digital recording device, such as a CCD array, limits the spatial resolution in images obtained by an optical imaging system, thereby degrading the image quality. Digital super-resolution (DSR) techniques are used to reconstruct a high-resolution (HR) image from multiple sub-pixel-shifted low-resolution images in order to improve the image quality. In this article, we formulate a mathematical framework for DSR using the generalized sampling theorem (GST). The GST-based DSR method's performance is evaluated by comparing it to existing resolution enhancement methods on the basis of evaluation metrics like percentage mean square error (%MSE), structural similarity index measure (SSIM), and peak signal-to-noise ratio (PSNR). The GST DSR method exhibits an overall superior quality image reconstruction based on quantitative analysis (with a near zero %MSE, SSIM around one, and improved PSNR in dB) and qualitative comparison. The robustness of the GST DSR method is further demonstrated in the presence of frame-to-frame shift estimation error using %MSE and SSIM by comparing it with multi-frame interpolation approaches. © 2024 Optica Publishing Group. All rights, including for text and data mining (TDM), Artificial Intelligence (AI) training, and similar technologies, are reserved.

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1. INTRODUCTION

Imaging systems commonly employ the techniques of super-resolution [1,2] in order to reconstruct a high-resolution (HR) image from low-resolution (LR) image frames. Extracting HR data from a single undersampled frame is difficult since it is an ill-posed problem. To circumvent this limitation, multiple undersampled image frames with sufficient diversity between them are used to reconstruct a HR image estimate, and this forms the basis of digital super-resolution (DSR) techniques.

In many conventional DSR methods, enhancement of spatial resolution mostly relies on interpolation-based methods [3,4] and have been employed in applications such as medical and satellite imaging. Carratto [5], Sajjad *et al.* [6], and Ling *et al.* [7] used interpolation as the basis of their DSR approaches. Other interpolation-based DSR techniques that have been explored by the community include edge-guided interpolation [8], interpolation utilizing multi-surface fitting [9], rational fractal interpolation [10], and new edge-directed interpolation [11]. Swapna *et al.* [12] performed DSR based on sparse representations of angiogram images. Bevilacqua and Roumy [13] employed linear-mapping-based DSR, whereas Shwetha and Rajathilagam [14] used DSR for breast cancer detection.

Sparse-representation-based DSR was employed on raw image patches by Yang *et al.* [15]. Many deep learning techniques have also been developed to accomplish DSR through single-frame interpolation [16–18]. Such techniques overcome the ill-posed nature of the DSR problem through training on large datasets [16], where the primary limitation is the need for large and reliable datasets to achieve robustness. Multi-frame interpolation-based methods [3] have been shown to perform well, but are highly sensitive to shift estimation error that can result in detrimental reconstruction performance. Performance can often be improved through application of additional constraints.

In this article, we develop a theoretical DSR framework using the generalized sampling theorem (GST) [19,20]. This technique is based on the assumption that the true image is band-limited, and for mathematical simplicity, assumes that the image is recorded by point-sized pixels. Since the pixel array captures undersampled data, the separation between two neighboring pixels is expressed as $\frac{L}{2B}$, $L \geq 2$, where B represents the band-limit of the true image in one-dimension (1D) and L denotes the undersampling factor. It is evident that the undersampling factor L dictates the number of undersampled

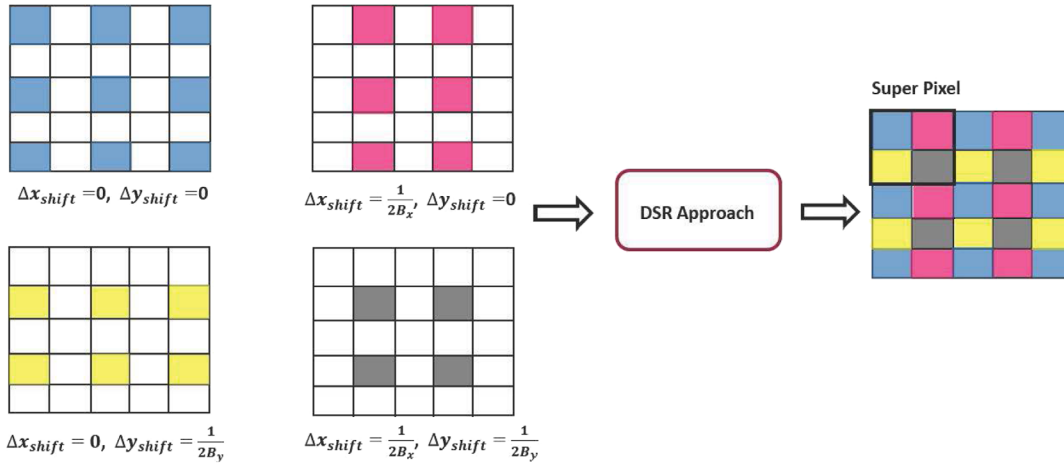


Fig. 1. Concept of the GST DSR method is shown for the trivial case of 1-pixel shifts along the horizontal, vertical, and diagonal directions. The undersampled data are extracted and stitched together based upon their respective shifts to obtain an estimate of the HR image.

image frames needed to reconstruct a HR image frame and the diversity between the individual frames is necessary for accuracy. There are many ways to acquire such diverse undersampled image frames. Some of these methods include sub-pixel translational shifts of the sensor array within the focal plane (phase diversity), sub-pixel longitudinal shifts of the sensor array along the defocus axis (defocus diversity), and tilts in the sensor array plane (tilt diversity). We have developed a mathematical framework based on the GST that allows for stitching these multiple undersampled frames together in the frequency domain [21]. Once complete, we can obtain the spatial HR image estimate by applying the inverse Fourier transform. An illustration of the GST DSR concept for the trivial case of LR image frames exhibiting whole pixel shifts is shown in Fig. 1. Given that it is impractical to achieve whole-pixel shifts in practice, we provide a more general development to allow for arbitrary sub-pixel shifts in our HR image model. We evaluate the performance of the GST DSR algorithm using both simulated and real data. We first analyze algorithm performance using synthetic data based upon Gaussian targets. We consider both simple scenes consisting of single spatial Gaussian distributions and more complex scenes composed of linear combinations of Gaussian distributions of varying widths and heights. We also employ real data taken from various datasets [16] in our evaluation. A comparative performance analysis using evaluation metrics such as percentage mean square error (%MSE) [16], structural similarity index measure (SSIM) [22], and peak signal-to-noise ratio (PSNR) [16] is conducted for the GST DSR approach and other conventional DSR techniques. The GST DSR method's robustness is further evaluated by examining the impact of inaccuracies in the shift estimates (as would be expected in real-world imaging scenarios) on the accuracy of the reconstructed HR image.

The paper is organized as follows: Section 2 discusses the mathematical formulation of the GST DSR algorithm. Section 3 describes application, evaluation, and discussion of the GST DSR and other relevant techniques to both simulated and real data. Finally, conclusions and future research directions are discussed in Section 4.

2. PROPOSED METHOD

A typical imaging system comprises a lens, a focal plane array (FPA) sensor, and the scene to be imaged. To provide a clear and detailed image, the FPA is positioned at the focal plane of the lens. In general, undersampling results due to the finite size of FPA pixels. In this study, we investigate a DSR method based upon the GST that makes use of phase diversity for HR image reconstruction. Multiple sub-pixel-shifted, undersampled image frames obtained via translational shifts of the FPA are used as input data for the method, and a HR image is produced as the output. Since translational shifts in the spatial domain correspond to a phase factor in the frequency domain, different amounts of shift contribute to phase diversity between different frames and hence this method is called a phase-diversity-based approach.

Consider a two-dimensional (2D) image $g(x, y)$ formed on the image plane of an imaging system that is band-limited in the range of $[(-B_x, B_x), (-B_y, B_y)]$. Such an image is recorded by a 2D FPA where we assume, for mathematical simplicity, that the pixels are point-sized. We consider the case when the FPA records undersampled data of $g(x, y)$ where the undersampling is quantified by the factor L and $\frac{1}{2B}$ denotes the pixel separation along each dimension. In a 2D image, the undersampling factors L_x and L_y , along the x and y axes, respectively, may be different in general, but here we assume that $L_x = L_y = L$.

As a proof of concept, a simple convincing argument can be made in favor of the phase diversity method in 1D. Let us examine a 1D image that is recorded by a FPA with under-sampling factor $L = 2$, resulting in a pixel spacing of $\frac{1}{B}$ for an image band-limited in the range of $(-B, B)$. Notice that the critically sampled HR data can be reconstructed in such a case by shifting the FPA by $\frac{1}{2B}$ within the focal plane to obtain an undersampled frame at the midway point between each pair of neighboring pixels. In practice, shifting the FPA by such an exact amount is challenging (and impractical), and hence there is a need for a method capable of stitching together multiple undersampled image frames that exhibit a more arbitrary frame-to-frame shift, i.e., where the shifts can be any fraction of the sampling interval

within the undersampled data space. The GST provides a mathematical framework to achieve this, as we have demonstrated in recent work [23].

Now, we consider the DSR approach in 2D where a set of four frames that exhibit no shift (reference) and shifts along the horizontal, vertical, and diagonal directions is employed. The 2D continuous image $g(x, y)$ has Fourier transform $G(v_x, v_y)$ and is assumed to be band-limited in the range of $[(-B_x, B_x), (-B_y, B_y)]$ with inverse Fourier transform

$$g(x, y) = \int_{-B_x}^{B_x} \int_{-B_y}^{B_y} G(v_x, v_y) e^{j2\pi(xv_x + yv_y)} dv_x dv_y. \quad (1)$$

For the case of undersampled images with an undersampling factor of $L_x = L_y = L = 2$, the above equation can be written as

$$\begin{aligned} g\left(\frac{mL_x}{2B_x}, \frac{nL_y}{2B_y}\right) &= g\left(\frac{m}{B_x}, \frac{n}{B_y}\right) = \int_{-B_x}^0 \int_{-B_y}^0 G(v_x, v_y) e^{j2\pi\left(\frac{mv_x}{B_x} + \frac{nv_y}{B_y}\right)} dv_x dv_y \\ &+ \int_{-B_x}^0 \int_0^{B_y} G(v_x, v_y) e^{j2\pi\left(\frac{mv_x}{B_x} + \frac{nv_y}{B_y}\right)} dv_x dv_y \\ &+ \int_0^{B_x} \int_{-B_y}^0 G(v_x, v_y) e^{j2\pi\left(\frac{mv_x}{B_x} + \frac{nv_y}{B_y}\right)} dv_x dv_y \\ &+ \int_0^{B_x} \int_0^{B_y} G(v_x, v_y) e^{j2\pi\left(\frac{mv_x}{B_x} + \frac{nv_y}{B_y}\right)} dv_x dv_y. \end{aligned} \quad (2)$$

The four integral terms in the above equation can be regrouped into a single integral term as

$$\begin{aligned} g\left(\frac{m}{B_x}, \frac{n}{B_y}\right) &= \int_{-B_x}^0 \int_{-B_y}^0 \left[G(v_x, v_y) + G(v_x, v_y + B_y) e^{j2\pi\left(\frac{mv_x}{B_x} + \frac{n(v_y + B_y)}{B_y}\right)} \right. \\ &+ G(v_x + B_x, v_y) e^{j2\pi\left(\frac{m(v_x + B_x)}{B_x} + \frac{nv_y}{B_y}\right)} \\ &\left. + G(v_x + B_x, v_y + B_y) e^{j2\pi\left(\frac{m(v_x + B_x)}{B_x} + \frac{n(v_y + B_y)}{B_y}\right)} \right] dv_x dv_y. \end{aligned} \quad (3)$$

Expanding the exponential terms, the above equation can be rewritten as

$$g\left(\frac{m}{B_x}, \frac{n}{B_y}\right) = \int_{-B_x}^0 \int_{-B_y}^0 J(v_x, v_y) e^{j2\pi\left(\frac{mv_x}{B_x} + \frac{nv_y}{B_y}\right)} dv_x dv_y, \quad (4)$$

where $J(v_x, v_y) \equiv [G(v_x, v_y) + G(v_x, v_y + B_y) + G(v_x + B_x, v_y) + G(v_x + B_x, v_y + B_y)]$. Now, consider a periodic function $J'(v_x, v_y)$ with a periodicity of $(\frac{2B_x}{L}, \frac{2B_y}{L})$, defined as

$$J'(v_x, v_y) = J(v_x, v_y) \forall (-B_x, -B_y) \leq (v_x, v_y) \leq (0, 0). \quad (5)$$

Since J' is a periodic function, it can be expressed in terms of a Fourier series such that

$$\begin{aligned} J'(v_x, v_y) &= \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \phi(m, n) e^{-j2\pi\left(\frac{Lmv_x}{2B_x} + \frac{Lnv_y}{2B_y}\right)}, \quad (m, n) \in I, \end{aligned} \quad (6)$$

where $\phi(m, n)$ is the Fourier series coefficient

$$\begin{aligned} \phi(m, n) &= \frac{L^2}{4B_x B_y} \int_{-B_x}^{-B_x + \frac{2B_x}{L}} \int_{-B_y}^{-B_y + \frac{2B_y}{L}} J'(v_x, v_y) \\ &\times e^{j2\pi\left(\frac{Lmv_x}{2B_x} + \frac{Lnv_y}{2B_y}\right)} dv_x dv_y. \end{aligned} \quad (7)$$

Substituting for $L = 2$, the above equation becomes

$$\phi(m, n) = \frac{1}{B_x B_y} \int_{-B_x}^0 \int_{-B_y}^0 J'(v_x, v_y) e^{j2\pi\left(\frac{mv_x}{B_x} + \frac{nv_y}{B_y}\right)} dv_x dv_y. \quad (8)$$

Since $J'(v_x, v_y) = J(v_x, v_y)$ for $(-B_x, -B_y) \leq (v_x, v_y) \leq (0, 0)$, upon comparing Eqs. (4) and (8) we obtain

$$\phi(m, n) = \frac{1}{B_x B_y} g\left(\frac{m}{B_x}, \frac{n}{B_y}\right). \quad (9)$$

From Eqs. (6) and (9), within the frequency range of $(-B_x, -B_y) \leq (v_x, v_y) \leq (0, 0)$,

$$J'(v_x, v_y) = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \frac{1}{B_x B_y} g\left(\frac{m}{B_x}, \frac{n}{B_y}\right) e^{-j2\pi\left(\frac{mv_x}{B_x} + \frac{nv_y}{B_y}\right)}. \quad (10)$$

$J'(v_x, v_y)$ can be replaced by $J(v_x, v_y) = [G(v_x, v_y) + G(v_x, v_y + B_y) + G(v_x + B_x, v_y) + G(v_x + B_x, v_y + B_y)]$ in Eq. (10), resulting in

$$\begin{aligned} &[G(v_x, v_y) + G(v_x + B_x, v_y) + G(v_x, v_y + B_y) \\ &+ G(v_x + B_x, v_y + B_y)] \\ &= \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \frac{1}{B_x B_y} g\left(\frac{m}{B_x}, \frac{n}{B_y}\right) e^{-j2\pi\left(\frac{mv_x}{B_x} + \frac{nv_y}{B_y}\right)}. \end{aligned} \quad (11)$$

The left-hand side (LHS) of Eq. (11) represents the summation of the values of the Fourier transform $G(v_x, v_y)$ at four different 2D frequencies (v_x, v_y) , $(v_x, v_y + B_y)$, $(v_x + B_x, v_y)$, and $(v_x + B_x, v_y + B_y)$, where each frequency lies in a different quadrant of the 2D frequency plane. The LHS of Eq. (11) contains all the unknowns of the system, while the right-hand side contains the measured undersampled data, which are all known quantities. For a given 2D frequency in the range $(-B_x, -B_y) \leq (v_x, v_y) \leq (0, 0)$, we can treat each of the terms, i.e., $G(v_x, v_y)$, $G(v_x + B_x, v_y)$, $G(v_x, v_y + B_y)$, and $G(v_x + B_x, v_y + B_y)$, as unknowns that require four independent equations in order to solve for them. Equation (11) thus provides the first of the four independent equations where the remaining three can be obtained using sub-pixel translational shifts in the xy image plane. The second frame is obtained

using a horizontal sub-pixel shift of Δx , the third frame with a vertical sub-pixel shift of Δy , and the fourth frame with a diagonal shift of Δx and Δy . We follow the analysis used to derive Eq. (11) to obtain similar equations for each shifted frame that we express in matrix form:

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ e^{j2\pi v_x \Delta x} & e^{j2\pi(v_x+B_x)\Delta x} & e^{j2\pi v_x \Delta x} & e^{j2\pi(v_x+B_x)\Delta x} \\ e^{j2\pi v_y \Delta y} & e^{j2\pi v_y \Delta y} & e^{j2\pi(v_y+B_y)\Delta y} & e^{j2\pi(v_y+B_y)\Delta y} \\ e^{j2\pi(v_x \Delta x + v_y \Delta y)} & e^{j2\pi((v_x+B_x)\Delta x + v_y \Delta y)} & e^{j2\pi(v_x \Delta x + (v_y+B_y)\Delta y)} & e^{j2\pi((v_x+B_x)\Delta x + (v_y+B_y)\Delta y)} \end{pmatrix} \begin{pmatrix} G(v_x, v_y) \\ G(v_x + B_x, v_y) \\ G(v_x, v_y + B_y) \\ G(v_x + B_x, v_y + B_y) \end{pmatrix} \\ = \begin{pmatrix} \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \frac{1}{B_x B_y} g\left(\frac{m}{B_x}, \frac{n}{B_y}\right) e^{-j2\pi\left(\frac{mv_x}{B_x} + \frac{nv_y}{B_y}\right)} \\ \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \frac{1}{B_x B_y} g\left(\frac{m}{B_x} + \Delta x, \frac{n}{B_y}\right) e^{-j2\pi\left(\frac{mv_x}{B_x} + \frac{nv_y}{B_y}\right)} \\ \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \frac{1}{B_x B_y} g\left(\frac{m}{B_x}, \frac{n}{B_y} + \Delta y\right) e^{-j2\pi\left(\frac{mv_x}{B_x} + \frac{nv_y}{B_y}\right)} \\ \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \frac{1}{B_x B_y} g\left(\frac{m}{B_x} + \Delta x, \frac{n}{B_y} + \Delta y\right) e^{-j2\pi\left(\frac{mv_x}{B_x} + \frac{nv_y}{B_y}\right)} \end{pmatrix}, \quad (12)$$

which can be expressed more concisely as

$$M_{\text{sys}} \bar{G} = \bar{D}. \quad (13)$$

Solving for $\bar{G}$ then yields

$$\bar{G} = M_{\text{sys}}^{-1} \bar{D}. \quad (14)$$

$\bar{G}$ is the unknown HR Fourier transform to be estimated, $\bar{D}$ are the known data vectors composed of the undersampled image frames, and M_{sys} represents the system matrix. For a given 2D frequency in the range $(-B_x, -B_y) \leq (v_x, v_y) \leq (0, 0)$, Eq. (14) can be used to estimate G across each of the four quadrants of the 2D frequency plane: $\bar{G}(1)$ provides the value in quadrant $(-B_x, -B_y) \leq (v_x, v_y) \leq (0, 0)$, $\bar{G}(2)$ in quadrant $(0, -B_y) \leq (v_x, v_y) \leq (B_x, 0)$, $\bar{G}(3)$ in quadrant $(-B_x, 0) \leq (v_x, v_y) \leq (0, B_y)$, and $\bar{G}(4)$ in quadrant $(0, 0) \leq (v_x, v_y) \leq (B_x, B_y)$. Once $\bar{G}$ is obtained, the high-resolution image estimate $\bar{g}(x, y)$ is obtained by taking the inverse Fourier transform.

Even though we have performed our analysis for the special case of $L = 2$, we can extend this work to larger values of L . Doing so introduces some challenges as the corresponding matrix equation becomes $\bar{G}_{L^2 \times 1} = M_{L^2 \times L^2}^{-1} \times \bar{D}_{L^2 \times 1}$, where the subscript represents the size of the corresponding matrix. Thus, theoretically, we can solve for larger values of L at an increased cost in computational complexity. This may be undesirable in practical applications and warrants further study. Moreover, as is evident from Eq. (13), the feasibility of the GST DSR method relies on the invertibility of the system matrix M_{sys} . M_{sys} depends on the spatial frequency (v_x, v_y) and the shifts $(\Delta x, \Delta y)$. For the case of $L = 2$, (v_x, v_y) vary in the range $(-B_x, -B_y) \leq (v_x, v_y) \leq (0, 0)$, so the user has to ensure that M_{sys} is invertible at each corresponding frequency. Clearly, for smaller values of L this is easier to manage, but as L increases this becomes more challenging.

3. RESULTS AND DISCUSSION

A. Application to Simulated Data

We first investigate algorithm performance by generating images of size 101×101 that consist of scenes containing

spatially distributed Gaussian functions. We use simple Gaussian functions of the forms below to generate symmetric (g_s), non-symmetric (g_b), and diagonally oriented (g_d) distributions according to

$$g_s(x, y) = e^{-\frac{x^2 + y^2}{\sigma^2}}, \quad (15)$$

$$g_b(x, y) = e^{-\frac{x^2}{\sigma_x^2} - \frac{y^2}{\sigma_y^2}}, \quad (16)$$

and

$$g_d(x, y) = e^{-\left(\frac{(x \cos \theta - y \sin \theta)^2}{\sigma_x^2} + \frac{(x \sin \theta + y \cos \theta)^2}{\sigma_y^2}\right)}. \quad (17)$$

We use the Gaussian distributions defined above to generate 12 different scenes. Scenes 1–5, denoted S1 through S5, consist of a single Gaussian distribution. Scenes S6 through S12 consist of linear combinations of Gaussian distributions of different variances using Eqs. (15)–(17) in order to create more spatially complex scenes [24]. The resulting simulated scenes are shown further below in Fig. 3. It is important to note that the GST DSR method requires the scenes to be band-limited. The Fourier transform of a Gaussian function is also Gaussian, and the fact that a Gaussian function tapers to zero can reasonably be assumed to be band-limited for all practical purposes. Here we assume that the band-limit is defined to be the frequency at which the magnitude of the Fourier transform is 0.005% of its peak value. The Fourier transform of the simulated scenes was computed to empirically determine the band-limit. For scenes S1 to S11, the estimated band-limit was $B = 0.1$ cycles/pixel-width and $B = 0.2$ cycles/pixel-width for S12.

For the purposes of applying the DSR GST algorithm, we assume that $L = 2$ and hence requires four undersampled frames for reconstruction. We assume that our generated scenes have a symmetric band-limit, i.e., $B_x = B_y = B$. We simulate four undersampled image frames using sub-pixel shifts $\leq \frac{1}{2B}$. We obtain these undersampled and shifted frames in two different ways. First, we use shift values of exactly $\frac{1}{2B}$ along the

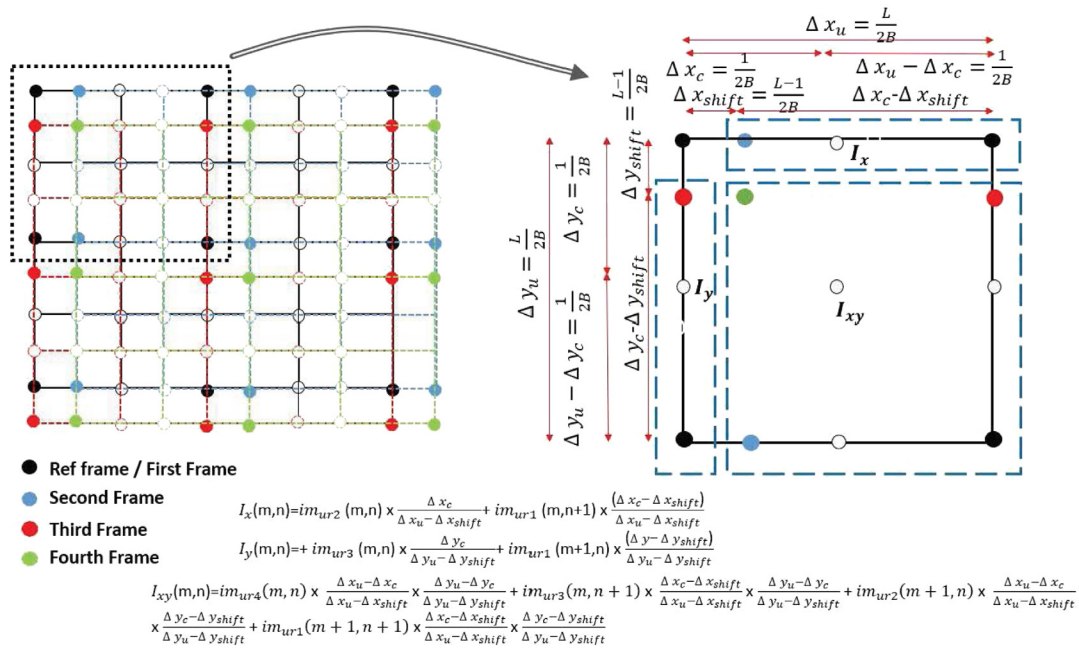


Fig. 2. Illustration of the interpolation process for estimating the HR scene from four sub-pixel-shifted LR data frames. The shifts include one along the horizontal axis, one along the vertical axis, and one along the diagonal axis. The mathematical expressions are provided that relate the HR and LR data frames.

horizontal, vertical, and diagonal directions to point-sample the high-resolution simulated ground truth scene, and refer to these images as the fixed shift (FS) case. Second, we similarly use random shift values that are $< \frac{1}{2B}$ along each direction and refer to these images as the random shift (RS) case. In real-world scenarios, where random shifts are more likely to be encountered, we felt it was important to study and compare results from each of these cases.

We apply the GST DSR technique to these data along with other existing enhancement techniques that include traditional bilinear and bicubic interpolation [16], super-resolution convolutional neural network (SRCNN) [16], and multi-frame interpolation (MFI) [3] (the concept behind MFI is illustrated in Fig. 2). MFI uses multiple frames for reconstruction whereas the other techniques are based upon a single-frame input with a shift of $\frac{1}{2B}$. Figure 3 displays the reconstruction results from each of these methods for each simulated scene along with the corresponding HR ground truth image. Tables 1 and 2 display %MSE [16] and SSIM [22] results for each reconstructed result against ground truth. In almost all cases, the GST DSR method exhibits lower %MSE for both the fixed and random shift cases against the other methods. SSIM values for GST DSR using either fixed or random shifts is nearly one for all images, indicating strong reconstruction performance. Hence, these results demonstrate accurate reconstruction for both simple and more complex simulated scenes.

B. Application to Real Data

We next apply the algorithm to data from real 2D datasets. Figure 4 shows example images and corresponding 201×201 subimages (indicated by the red box) that were extracted for our analysis; 125 images in total were used that were selected from

Set5, Set14, and the BSD200 dataset in [16] and example demo datasets from the MATLAB platform. The GST DSR technique was applied to undersampled images obtained similarly as described in the previous section. The estimated band-limit along each dimension of these images was $B = 0.5$ cycles/pixel-width. The same set of algorithms from the previous section was applied to these data to estimate the HR scene with the results for several images shown in Fig. 5. Tables 3 and 4 display mean PSNR and SSIM results of each reconstructed image against ground truth averaged across all 125 selected images. Tables 5 and 6 display the PSNR [16] and SSIM for some selected images from the provided datasets. The findings indicate that the GST DSR method, using both fixed and random shifts, demonstrates strong performance similar to results achieved on the simulated scenes of the previous section. The GST DSR method consistently yields higher performance against the other evaluated methods, as is particularly apparent in the significantly higher PSNR scores and SSIM values that are near one.

C. Performance Evaluation of GST DSR Method

In practical applications, it is frequently challenging to precisely control the motion of sensor platforms, necessitating the estimation of the frame-to-frame shift from the data. Such motion-estimation algorithms are prone to error in the estimated shifts. As a result, the quality of high-resolution scene reconstruction is influenced by these shift estimation errors. Here we examine how shift estimation error impacts the reconstruction accuracy of the GST DSR and MFI methods. Let us denote the true value of shift along the x -axis as Δx_t and the percentage error in shift estimation as $\% \Delta x$. The estimated value of the shift can then be expressed as

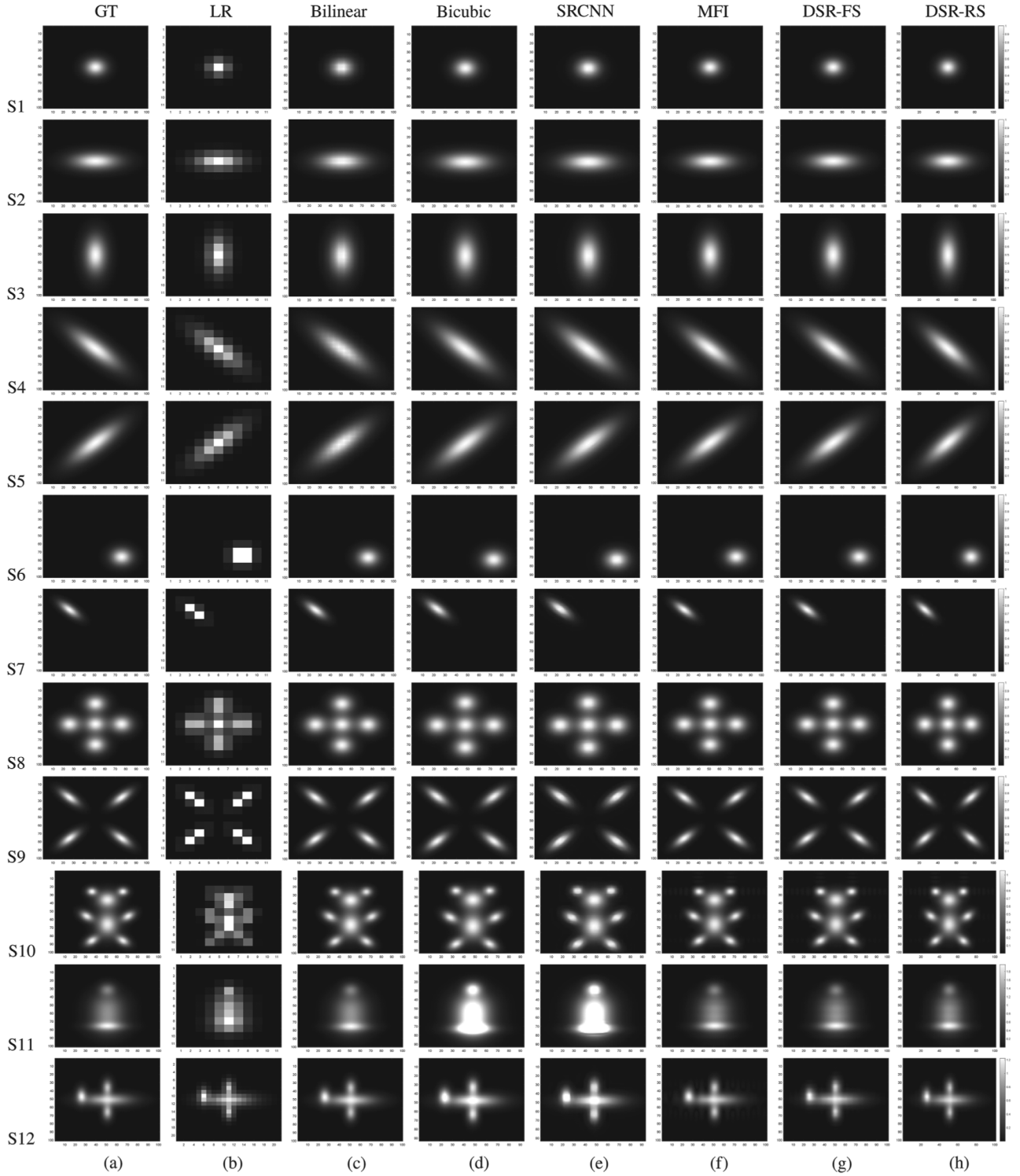


Fig. 3. Simulated scene reconstruction results: (a) ground truth, (b) LR image, (c) bilinear interpolation [16], (d) bicubic interpolation [16], (e) SRCNN [16], (f) MFI [3], (g) DSR FS, and (h) DSR-RS.

$$\Delta \hat{x} = \Delta x_t \left(1 + \frac{\% \Delta x}{100} \right). \quad (18)$$

The error in HR reconstruction using the GST DSR and MFI methods evaluated using %MSE and SSIM as a function of percentage shift estimation error is shown in Figs. 6 and 7 for six selected images, three simulated scenes (S1, S5, and S10), and three real images selected from Set5 dataset in [16] (denoted

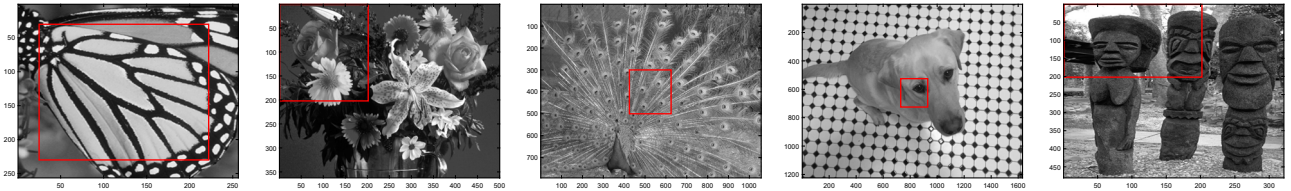
R1, R2, and R3). The plots indicate that the %MSE and SSIM are near zero and one, respectively, for both methods at 0% shift estimation error. As the error in shift estimation increases, the %MSE gradually increases, and the SSIM value gradually decreases for both methods, but the rate of degradation in reconstruction accuracy is slower for the GST DSR method as compared to the MFI method. A visual comparison of ground

Table 1. %MSE Evaluation of Different Enhancement Methods against Ground Truth for the Simulated Scenes

2D Data	Bilinear [16]	Bicubic [16]	SRCNN [16]	MFI [3]	DSR FS	DSR RS
S1	0.0172	4.7e-04	5.73e-04	6.91e-10	8.12e-10	2.16e-09
S2	0.0178	8e-04	6.8e-04	1.026e-05	9.44e-07	1.411e-06
S3	0.0178	8e-04	6.79e-04	1.026e-05	9.44e-07	6.47e-06
S4	0.0214	1.7e-04	4.32e-04	2.561e-05	1.824e-06	4.012e-06
S5	0.0214	5.3e-04	7.38e-04	2.51e-05	2.148e-06	4.148e-06
S6	0.0172	4.8e-04	5.5e-04	1.77e-06	1.872e-07	5.95e-07
S7	0.0026	3.4e-04	1.37e-04	1.282e-04	1.285e-04	2.020e-04
S8	0.0033	2.24e-04	2.03e-04	1.735e-06	1.96e-07	5.59e-07
S9	0.0102	1.416e-03	5.9e-03	3.88e-04	4.113e-04	7.02e-04
S10	0.0132	0.003	0.008	0.0045	0.0047	0.005
S11	0.0014	0.004	0.026	8.17e-03	8.14e-04	1.6e-04
S12	0.06	0.032	0.0118	0.010	0.015	0.006

Table 2. SSIM Evaluation of Different Enhancement Methods against Ground Truth for the Simulated Scenes

2D Data	Bilinear [16]	Bicubic [16]	SRCNN [16]	MFI [3]	DSR FS	DSR RS
S1	0.989	0.990	0.996	1	1	1
S2	0.987	0.990	0.995	0.999	1	1
S3	0.987	0.990	0.995	0.999	1	0.999
S4	0.984	0.995	0.996	0.998	1	0.999
S5	0.989	0.990	0.996	1	1	1
S6	0.989	0.990	0.997	0.999	1	1
S7	0.990	0.995	0.995	0.998	0.9981	0.998
S8	0.980	0.990	0.993	0.999	1	1
S9	0.98	0.98	0.98	0.99	0.995	0.995
S10	0.98	0.90	0.986	0.96	0.99	0.99
S11	0.982	0.99	0.9906	0.991	0.992	0.9906
S12	0.982	0.987	0.98	0.996	0.997	0.996

**Fig. 4.** Selected real images from various datasets in [16]. The red box indicates selected subimages that were extracted for our study; 125 images in total were used.

truth data with reconstructed data in the presence of 20% shift estimation error is shown in Fig. 8, clearly demonstrating that the GST DSR method is more robust to shift estimation error.

Although the other methods we evaluated exhibit lower estimation accuracy, their computational time (shown in Table 7) is lower than that of GST DSR, making them acceptable in situations where computational efficiency must be prioritized over higher accuracy. Furthermore, one shortcoming of the GST DSR approach is the need to generate a large number of low-resolution data frames in addition to requiring prior knowledge of the band-limit in practical scenarios.

4. CONCLUSION

In this work we presented a multi-frame DSR approach based upon the GST. We applied the algorithm to reconstruct high-resolution scene estimates using both simulated and real data.

We evaluated algorithm performance on these data along with other DSR and interpolation methods and found that the DSR GST algorithm could obtain relatively superior performance, consistently achieving near-zero values of %MSE, SSIM scores near one, and high values of PSNR. Furthermore, we studied the effect of motion estimation error on reconstruction accuracy of both the DSR GST and MFI approaches. We found the GST DSR method to be quite robust to shift estimation error whereas the MFI performance suffered greatly. This was apparent both visually in reconstructed images as well as in obtained %MSE and SSIM results when compared with HR ground truth images. One limitation of the technique is that it requires prior knowledge of the input data band-limit and furthermore would require a large number low-resolution image frames with sufficient phase diversity for large values of the undersampling factor L .

The work can be expanded to incorporate tilt and defocus diversity in place of phase diversity or a combination of the three

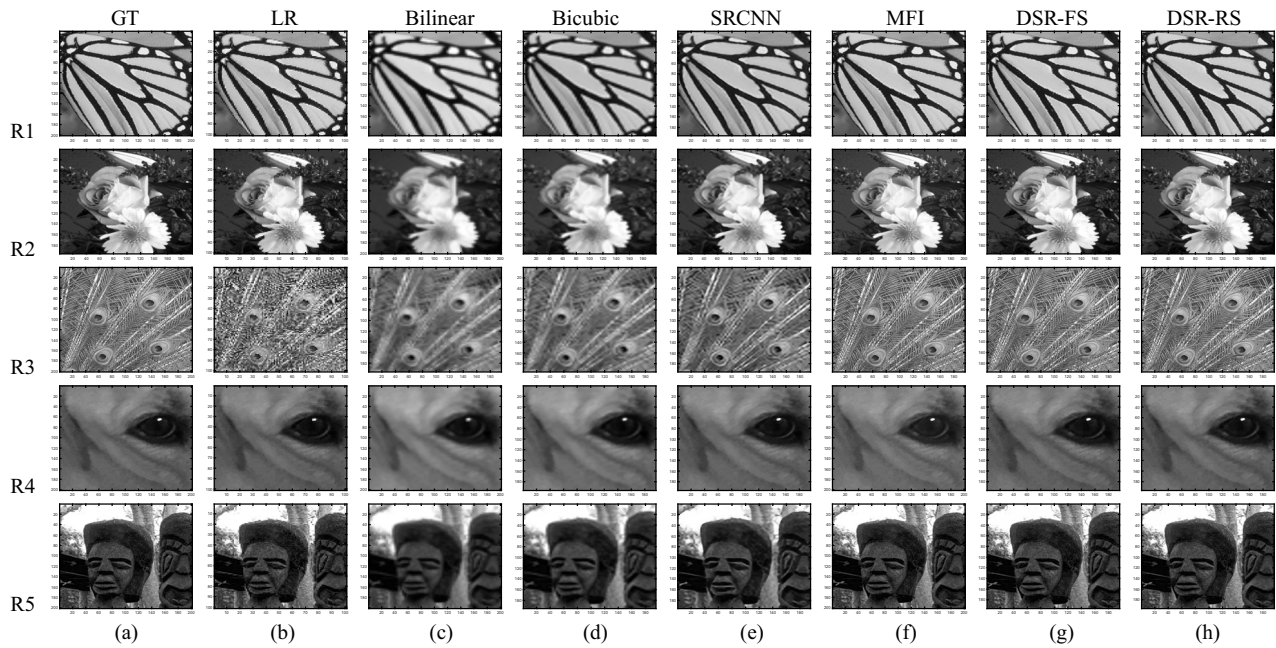


Fig. 5. Reconstruction results from real image data: (a) ground truth, (b) LR image, (c) bilinear interpolation [16], (d) bicubic interpolation [16], (e) SRCNN [16], (f) MFI [3], (g) DSR FS, and (h) DSR-RS.

Table 3. Mean PSNR Scores of Different Enhancement Methods against Ground Truth for Real Data from the Indicated Datasets

2D Data	Bilinear [16]	Bicubic [16]	SRCNN [16]	MFI [3]	DSR FS	DSR RS
Set5 [16]	29.57	29.71	32.52	32.55	52.42	52.16
Set14 [16]	27.9	27.10	28.71	32.03	53.94	50.73
BSD200 [16]	27.12	27.22	28.47	33.08	52.43	51.5
MATLAB	30.64	30.91	33.22	32.42	58.35	57.52

Table 4. Mean SSIM Scores of Different Enhancement Methods against Ground Truth for Real Data from the Indicated Datasets

2D Data	Bilinear [16]	Bicubic [16]	SRCNN [16]	MFI [3]	DSR FS	DSR RS
Set5 [16]	0.835	0.868	0.907	0.970	0.998	0.997
Set14 [16]	0.702	0.746	0.793	0.970	0.997	0.988
BSD200 [16]	0.718	0.757	0.803	0.972	0.997	0.992
MATLAB	0.736	0.766	0.800	0.965	0.997	0.997

Table 5. Performance Comparison of Different Enhancement Methods with DSR Approach Based on PSNR (in dB)

2D Data	Bilinear [16]	Bicubic [16]	SRCNN [16]	MFI [3]	DSR FS	DSR RS
R1	23.12	23.32	28.09	31.79	50.69	50.4
R2	28.28	25.29	27.39	36.17	57.99	56
R3	15	15.01	15.20	32.30	55.34	54
R4	40.42	38.56	41.31	40	53.41	52.7
R5	27.30	23.95	24.97	36.93	43.54	43

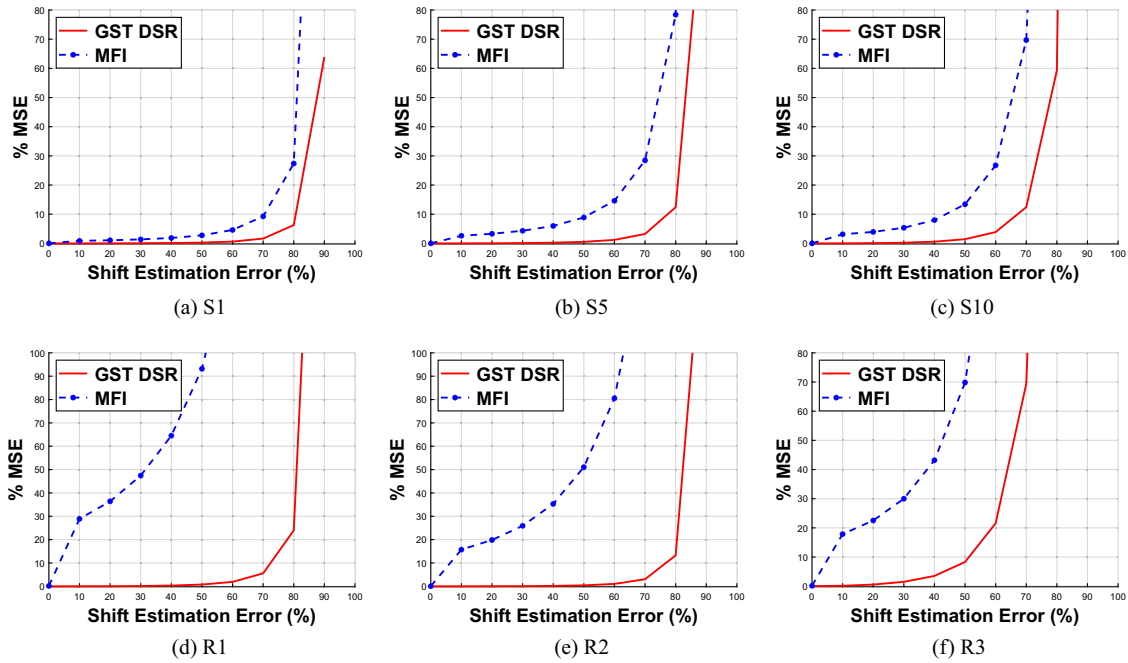
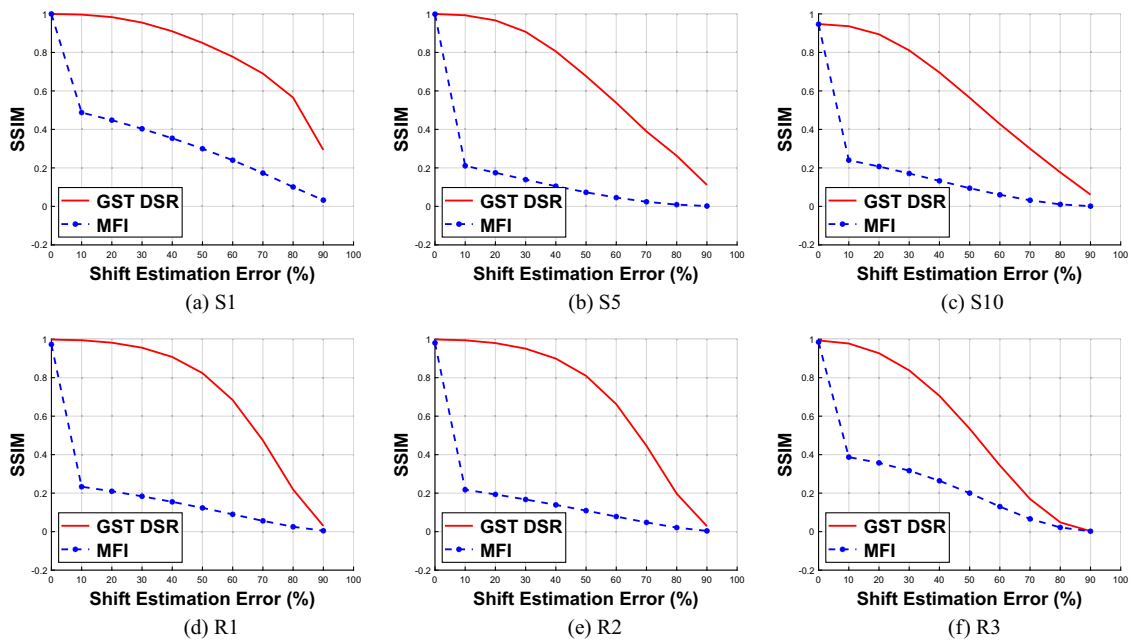
for resolution improvement of high undersampling factors. Defocus diversity can be achieved by moving the FPA along the principal axis, capturing different degrees of defocus blur. Tilt diversity can be achieved by tilting the FPA by various amounts with respect to the principal axis. We next plan to apply the technique to imagery from division of focal plane imaging polarimeter data. Such imaging systems are ideal candidates for

DSR techniques due to the inherent downsampling that results from pixel-to-pixel masking with mosaics of linear polarizers in these devices, resulting in both aliasing and instantaneous-field-of-view (IFOV) error that manifests in estimated Stokes vector imagery [25–27].

Disclosures. The authors declare no conflicts of interest.

Table 6. Performance Comparison of Different Enhancement Methods with DSR Approach Based on SSIM Measure

2D Data	Bilinear [16]	Bicubic [16]	SRCNN [16]	MFI [3]	DSR FS	DSR RS
R1	0.768	0.814	0.9117	0.973	0.998	0.997
R2	0.711	0.753	0.820	0.987	0.998	0.998
R3	0.156	0.191	0.262	0.985	0.9936	0.992
R4	0.929	0.950	0.963	0.960	1	1
R5	0.591	0.641	0.700	0.982	0.997	0.996

**Fig. 6.** Comparison of %MSE with shift estimation error for both DSR and MFI methods using simulated and real targets. (a) S1, (b) S5, (c) S10, (d) R1, (e) R2, and (f) R3.**Fig. 7.** Comparison of SSIM with shift estimation error for both DSR and MFI methods using simulated and real targets. (a) S1, (b) S5, (c) S10, (d) R1, (e) R2, and (f) R3.

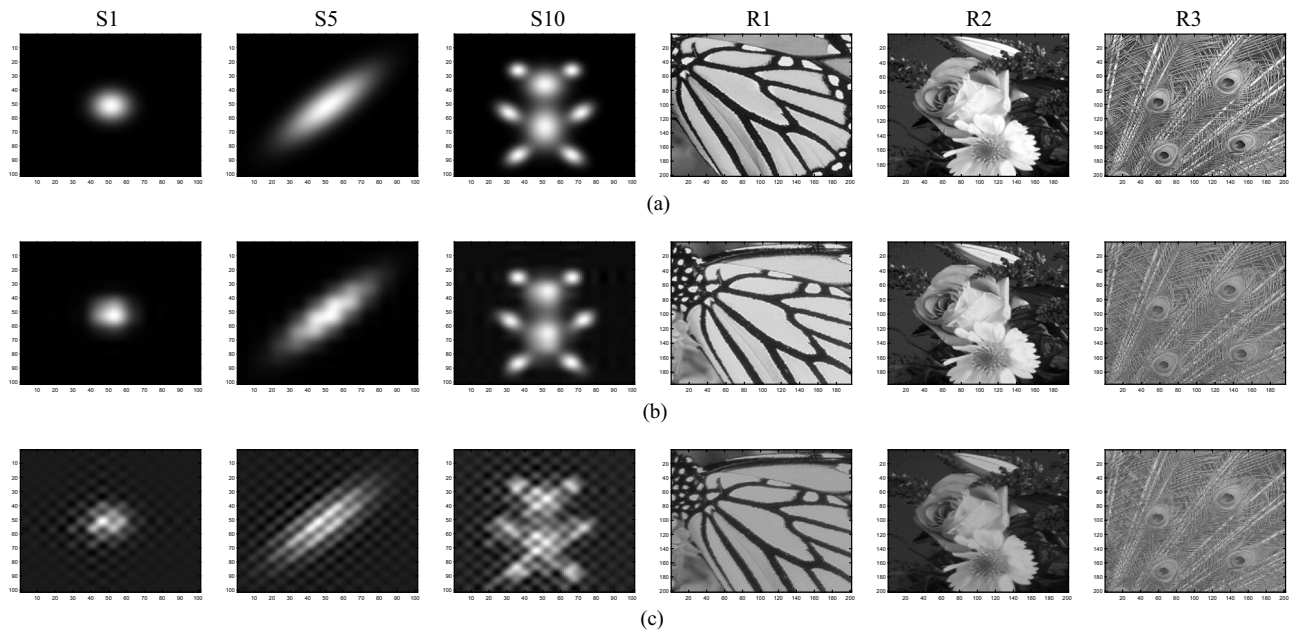


Fig. 8. Visual comparison of ground truth and HR reconstructed data for 20% shift error. (a) Ground truth. (b) DSR method. (c) MFI method.

Table 7. Comparison of Computational Time of GST DSR with the Existing Methods

	Bilinear	Bicubic	SRCNN	MFI	GST DSR
Running time (in s)	7.65	7.41	6.36	5.57	7.33

Data availability. Data underlying the results presented in this paper are available in [16] and in the MATLAB platform [1–22,24–29].

Supplemental document. See Supplement 1 for supporting content.

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